2 Two Kinds of Reasoning

Dmytro Zinkevych/Alamy Stock Photo

Time to look more closely at arguments—the kind that actually show something (unlike the red herrings, emotional appeals, and other fallacies we talk about later).

Students will learn to . . .

1. **Recognize general features of arguments**
2. **Distinguish between deductive and inductive arguments and evaluate them for validity, soundness, strength, and weakness**
3. **Identify unstated premises**
4. **Identify a balance of considerations argument and an inference to the best explanation (IBE)**
5. **Distinguish between ethos, pathos, and logos as means of persuasion**
6. **Use techniques for understanding and evaluating the structure and content of arguments**

ARGUMENTS: GENERAL FEATURES

To repeat, an argument consists of two parts. One part, the premise, is intended to provide a reason for accepting the second part, the conclusion. This statement is *not* an argument:

> God exists.

It's just a statement.

Likewise, *this* is not an argument:

> God exists. That's as plain as the nose on your face.

It's just a slightly more emphatic statement.

Nor is this an argument:

> God exists, and if you don't believe it, you will go to hell.

It just tries to scare us into believing God exists.

Also not an argument:

> I think God exists, because I was raised a Baptist.

Yes, it looks a bit like an argument, but it isn't. It merely explains why I believe in God.

On the other hand, this *is* an argument:

> God exists because something had to cause the universe.

The difference between this and the earlier examples? This example has a premise ("something had to cause the universe") and a conclusion ("God exists").

As we explained in Chapter 1 (see page 9), an argument always has two parts: a premise part and a conclusion part. The premise part is intended to give a reason for accepting the conclusion part.

This probably seems fairly straightforward, but one or two complications are worth noting.

Conclusions Used as Premises

The same statement can be the conclusion of one argument and a premise in another argument:

> **Premise:** The brakes aren't working, the engine burns oil, the transmission needs work, and the car is hard to start.
> **Conclusion 1:** The car has outlived its usefulness.
> **Conclusion 2:** We should get a new car.

In this example, the statement "The car has outlived its usefulness" is the conclusion of one argument, and it is also a premise in the argument for the claim that we should get a new car.

Clearly, if a premise in an argument is uncertain or controversial or has been challenged, you might want to defend it—that is, argue that it is true. When you do, the premise becomes the conclusion of a new argument. However, every chain of reasoning must begin somewhere. If we ask a speaker to defend each premise with a further argument, and each premise in that argument with a further argument, and so on and so on, we eventually find ourselves being unreasonable, much like four-year-olds who keep asking "Why?" until they become exasperating. If we ask a speaker why he thinks the car has outlived its usefulness, he may mention that the car is hard to start. If we ask him why he thinks the car is hard to start, he probably won't know what to say.

Unstated Premises and Conclusions

Another complication is that arguments can contain unstated premises. For example:

> **Premise:** You can't check out books from the library without an ID.
> **Conclusion:** Bill won't be able to check out any books.

The unstated premise must be that Bill has no ID.

An argument can even have an unstated conclusion. Here is an example:

> The political party that best reflects mainstream opinion will win the presidency in 2024 and the Republican Party best reflects mainstream opinion.

If a person said this, he or she would be implying that the Republican Party will win the presidency in 2024; that would be the unstated conclusion of the argument.

Unstated premises are common in real life because sometimes they seem too obvious to need mentioning. The argument "the car is beyond fixing, so we should get rid of it" actually has an unstated premise to the effect that we should get rid of any car that is beyond fixing; but this may seem so obvious to us that we don't bother stating it.

Unstated conclusions also are not uncommon, though they are less common than unstated premises.

We'll return to this subject in a moment.

Conclusion Indicators

when the words in the following list are used in arguments, they usually indicate that a premise has just been offered and that a conclusion is about to be presented. (The three dots represent the claim that is the conclusion.)

Thus . . .	Consequently . . .
Therefore . . .	So . . .
Hence . . .	Accordingly . . .
This shows that . . .	This implies that . . .
This suggests that . . .	This proves that . . .

Example:

> Stacy drives a Porsche. This suggests that either she is rich or her parents are.

The conclusion is

> Either she is rich or her parents are.

The premise is

> Stacy drives a Porsche.

TWO KINDS OF ARGUMENTS

Good arguments come in two varieties: deductive demonstrations and inductive supporting arguments.

Deductive Arguments

The premise (or premises) of a good *deductive* argument, if true, *proves or demonstrates* (these being the same thing for our purposes) its conclusion. However, there is more to this than meets the eye, and we must begin with the fundamental concept of deductive logic, *validity.* An argument is **valid** *if it isn't possible for the premise (or premises) to be*

Premise Indicators

when the words in the following list are used in arguments, they generally introduce premises. They often occur just *after* a conclusion has been given. A premise would replace the three dots in an actual argument.

Since . . .
f or . . .
in view of . . .
This is implied by . . .

Example:

Either Stacy is rich or her parents are, since she drives a Porsche.

The premise is the claim that Stacy drives a Porsche; the conclusion is the claim that either Stacy is rich or her parents are.

true and the conclusion false. This may sound complicated, but it really isn't. An example of a valid argument will help:

Premises: Jimmy Carter was president immediately before Bill Clinton, and George W. Bush was president immediately after Bill Clinton.
Conclusion: Jimmy Carter was president before George W. Bush.

As you can see, it's impossible for these premises to be true and this conclusion to be false. So the argument is valid.

However, you may have noticed that the premises contain a mistake. Jimmy Carter was not president immediately before Bill Clinton. George H. W. Bush was president immediately before Bill Clinton. Nevertheless, even though a premise of the preceding argument is not true, the argument is still valid, because it isn't possible for the premises to be true and the conclusion false. Another way to say this: If the premises *were* true, the conclusion *could not* be false—and that's what "valid" means.

Now, when the premises of a valid argument *are* true, there is a word for it. In that case, the argument is said to be **sound.** Here is an example of a sound argument:

Premises: Bill Clinton is taller than George W. Bush, and Jimmy Carter is shorter than George W. Bush.
Conclusion: Therefore, Bill Clinton is taller than Jimmy Carter.

This argument is sound because it is valid and the premises are true. As you can see, if an argument is sound, then its conclusion has been demonstrated.

Inductive Arguments

Again, the premise of a good deductive argument, if true, demonstrates that the conclusion is true. This brings us to the second kind of argument, the *inductive* argument. The premise of a good *inductive* argument doesn't demonstrate its conclusion; it *supports* it.

It supports it in the sense that it raises the probability of the conclusion from what it would be if it (the premise) weren't true. For example:

After 2 P.M. the traffic slows to a crawl on the Bay Bridge.
Therefore, it probably does the same thing on the Golden Gate Bridge.

The fact that traffic slows to a crawl after 2 P.M. on the Bay Bridge does not demonstrate or prove that it does that on the Golden Gate Bridge; it *supports* that conclusion. It makes it somewhat more likely that traffic on the Golden Gate Bridge slows to a crawl after 2 P.M.

Here is another example of an inductive argument:

Nobody has ever run a mile in less than three minutes.
Therefore, nobody will ever run a mile in less than three minutes.

Like the first argument, the premise supports the conclusion but does not demonstrate or prove it. The premise supports the conclusion in the sense that the fact that nobody has ever run a mile in less than three minutes makes it more probable that nobody will ever run a mile in less than three minutes.

If you are thinking that support is a matter of degree and that it can vary from just a little to a whole lot, you are right. Thus, inductive arguments are better or worse on a scale, depending on how much support their premises provide for the conclusion. Logicians have a technical word to describe this situation. The more support the premise of an inductive argument provides for the conclusion, the **stronger** the argument; the less support it provides, the **weaker** the argument. Put another way, one argument for a conclusion is weaker than another if it fails to raise the probability of the conclusion by as much. Thus, the first argument given below is weaker than the argument that follows it.

After 2 P.M. the traffic slows to a crawl on the Bay Bridge.
Therefore, it probably does the same on the Golden Gate Bridge.

After 2 P.M. the traffic slows to a crawl on the Bay Bridge, the San Mateo Bridge, the San Rafael Bridge, and the Dumbarton Bridge.
Therefore, it probably does the same thing on the Golden Gate Bridge.

The second argument is stronger than the first argument because its premise makes its conclusion more likely. The more bridges in a region on which traffic slows at a given time, the more likely it is that that phenomenon is universal on the bridges in the region.

One more example of an inductive argument:

Alexandra rarely returns texts.
Therefore, she probably rarely returns emails.

Once again, the premise does not demonstrate or prove the conclusion, but supports it in the sense of raising its probability. The differences between texting and emailing are sufficiently significant that the premise does not offer a great deal of support for the conclusion, but it does offer some. If Alexandra rarely returned telephone calls or letters as well as texts, that would make the argument stronger.

In Chapter 11, we will explain the criteria for evaluating inductive arguments.

BEYOND A REASONABLE DOUBT

In common law, the highest standard of proof is proof "beyond a reasonable doubt." If you are a juror in a criminal trial, evidence will be presented to the court—facts that the interested parties consider relevant to the crime. Additionally, the prosecutor and counsel for the defense will offer arguments connecting the evidence to (or disconnecting it from) the guilt or innocence of the defendant. When the jury is asked to return a verdict, the judge will tell the jury that the defendant must be found not guilty unless the evidence proves guilt *beyond a reasonable doubt.*

Proof beyond a reasonable doubt actually is a lower standard than deductive demonstration. Deductive demonstration corresponds more to what, in ordinary English, might be expressed by the phrase "beyond any *possible* doubt." Recall that in logic, a proposition has been demonstrated when it has been shown to be the conclusion of a sound argument—an argument in which (1) all premises are true and (2) it is impossible for the premises to be true and for the conclusion to be false. In this sense, many propositions people describe as having been demonstrated or proved, such as that smoking causes lung cancer or that the DNA found at a crime scene was the defendant's, have not actually been proved in the logician's sense of the word. So, in real life, when people say something has been demonstrated, they may well be speaking informally. They may not mean that something is the conclusion of a sound deductive argument. However, when we—the authors—say that something has been demonstrated, that is *exactly* what we mean.

TWO KINDS OF DEDUCTIVE ARGUMENTS

When Parker flies, he flies Economy Class. When Moore flies, so does he. Their editor, however, luxuriates in Business Class. Some people, such as perhaps George and Amal Clooney, fly First Class.

"Economy Class," "Business Class," and "First Class" are names of categories. In fact, every noun or noun phrase is the name of a category, because a category is just a class or classification of things. You could also say that a category is a population. "Frequent flyers," "people," "woodchucks," "Robert E. Lee"—these words all denote or name categories. ("Robert E. Lee" names a category with one thing in it, Robert E. Lee.)

Categorical logic is the logic of categorical arguments, arguments that relate categories. We take a close look at the logic of categorical arguments in Chapter 9. Categorical arguments are one of the two kinds of deductive argument we examine in this book. Here is an example of a categorical argument, dealing with the categories "movie stars," "wealthy people," and "people who fly Economy Class."

> All movie stars are wealthy, and no wealthy people fly Economy Class. Therefore, no movie stars fly Economy Class.

The second kind of deductive argument we consider relates propositions which can be or have been compounded from simpler propositions by means of logical "operators" such as "not," "and," "or," and "if...then..." This kind of deductive argument is the subject of what is known as "**sentential**" or "**propositional**" or "**truth-functional**" **logic,** which we take up in Chapter 10. An example of a truth-functional argument is this.

> If Jones was in Toledo then Jackson was not in Peoria. In addition, either Jackson was in Peoria or Smith was in Cleveland. Therefore, if Smith was not in Cleveland, then Jones was not in Toledo.

Propositions like those used in this argument are called "truth-functional" because the truth or falsity ("truth value") of a compound proposition, such as "If Smith was not in Cleveland then Jones was not in Toledo" derives from (or is a "function" of) the truth or falsity of the simpler propositions "Smith was not in Cleveland" and "Jones was not in Toledo." And these last two propositions derive their truth or falsity from the truth or falsity of the even simpler propositions "Smith was in Cleveland" and "Jones was in Toledo."

FOUR KINDS OF INDUCTIVE ARGUMENTS

In Chapter 10 we examine four important kinds of inductive argument. First, is **Generalizing from a Sample,** an argument in which you inductively conclude that all or most or some percentage of all the members of a population have an attribute because all or most or some percentage of the members of a sample of the population have that attribute. An example is this.

> I've liked the majority of Professor Moore's lectures so far; therefore, probably I will like the majority of all of Professor Moore's lectures.

The second kind of inductive argument is the **Statistical Syllogism,** which we prefer to call **De-generalizing** or Reverse Generalizing or Instantiating. This type of argument is the reverse of generalizing from a sample. In it, you inductively conclude that particular members of a population have an attribute because some high proportions of all the population's members have that attribute.

> Most days that are hot and humid are days with thunderstorms. It is hot and humid today. Therefore, probably there will be a thunderstorm today.

Here the "population" is the population of hot and humid days, most members of which are said to have the attribute of having thunderstorms.*

In the next and third kind of inductive argument we cover in this book, a conclusion is derived from an analogy. More precisely, an **Argument from Analogy** is an inductive argument that something has an attribute because a similar thing has that attribute. As we will see in Chapter 11, analogical reasoning is foundational in law, ethics, science, and the discipline of History. Here is an example of an argument from analogy:

> Forcing Americans to buy health insurance for their own good is like forcing them to eat broccoli for their own good. It would be wrong to force people to eat broccoli for their own good; therefore, it would be wrong to force them to buy health insurance for their own good.**

*The premise of a de-generalizing argument is often couched in language that says that one thing is a "sign" of another. Here is the argument just given, re-packaged as a so-called "sign argument":

> *Heat and humidity are signs of impending thunderstorms. It is hot and humid today. Therefore, probably there will be a thunderstorm today.*

**Many so-called "sign-arguments" can also be analyzed as analogical arguments. Here is an example of a sign-argument that can be analyzed as an argument from analogy.

> *When Sparky scratches like that it's a sign he has fleas. Therefore he has fleas.*

You can see that we might analyze this as an analogy between this occasion on which Sparky has been scratching and past occasions on which he has been scratching:

> *This occasion is like past occasions on which Sparky has been scratching. On those past occasions he has had fleas. Therefore on this occasion he has fleas.*

if you thought you could also analyze the very same argument as a case of de-generalizing, you would be right. The moral is that so-called sign arguments need not be regarded as a separate kind of inductive argument.

Finally, the last type of inductive argument covered in this book is a cause-and-effect or (for short) a causal argument. **Causal Arguments** inductively support a claim that asserts or implies cause-and-effect, or, alternatively, use such a cause-and-effect claim as a premise in an argument to establish that something happened or is the case. Let's look at examples:

First, sometimes we focus on an event and derive a conclusion about what its probable effect is:

> The toilet is leaking. It isn't possible for the toilet to leak without damaging the floor. Therefore the floor is damaged.

Other times we focus on an event and derive a conclusion about what its probable cause must have been, as in this example.

> There is water on the floor. The only way water could get on the floor is from a leaking toilet. Therefore, the leaking toilet caused the water on the floor.

Finally, in a variation of this type of argument known as **Inference to the Best Explanation,** a statement that asserts or implies cause-and-effect is used as a *premise* in an argument intended to show that something is the case. An example:

> There is water on the floor. The most likely cause is a leaking toilet. Therefore, the toilet is leaking.

As can be seen, in an "inference to the best explanation" the cause-and-effect claim is not the conclusion, but a premise. This kind of argument in effect concludes that something is the case because it's the best (most likely) explanation of something else that we are interested in.

Some logic books discuss other kinds of inductive arguments, including sign arguments and arguments from examples. Usually these can be analyzed as instances of one or another of the four kinds of inductive arguments mentioned above, all of which we examine in Chapter 11.

TELLING THE DIFFERENCE BETWEEN DEDUCTIVE AND INDUCTIVE ARGUMENTS

A useful strategy for telling the difference between deductive and inductive arguments is to memorize a good example of each kind. Here are good examples of each:

> **Valid Deductive Argument:** Juan lives on the equator. Therefore, Juan lives midway between the North and South poles.

> **Relatively Strong Inductive Argument:** Juan lives on the equator. Therefore, Juan lives in a warm climate.

Study the two examples so that you understand the difference between them. In the left example, if you know the definition of "equator," you already know it is midway between the poles. The right example is radically different. The definition of "equator" does not contain the information that it is warm. So:

> If the conclusion of an argument is true *by definition* given the premise or premises, it is a valid deductive argument.

Often it is said that a valid deductive argument is valid due to its "form." Thus, consider this argument:

> If Juan is a fragglemop, then Juan is a snipette. Juan is not a snipette. Therefore, Juan is not a fragglemop.

What makes this argument valid is its form:

> If P then Q.
> Not-Q.
> Therefore not-P.

You can see, however, that ultimately what makes the argument valid, and makes its form a valid form, is the way the words "If . . . then" and "not" work. If you know the way those words work, then you already know that the conclusion must be true given the two premises.

Another way of telling the difference between a deductive argument and an inductive argument is this: You generally would not say of a deductive argument that it supports or provides evidence for its conclusion. It would be odd to say that Juan's living on the equator is *evidence* that he lives midway between the poles, or that it *supports* that claim. Thus:

> If it sounds odd to speak of the argument as providing evidence or support for a contention, that's an indication it is a deductive argument.

It would sound very odd to say, "The fact Sparky is a dog is evidence he is a mammal." Sparky's being a dog isn't *evidence* he is a mammal: it's *proof.* "Sparky is a dog; therefore he is a mammal" is a valid deductive argument.

Finally, an argument with a subjective judgment as its conclusion is not inductive. Inductive arguments are stronger or weaker on a scale, and one inductive argument for a conclusion is stronger than another if it raises the probability of the conclusion by a greater amount. Because (as explained in Chapter 1) degrees of probability do not attach to subjective judgments, you can't think of an argument with a subjective judgment as relatively strong or week.*

*we can imagine someone saying, "i just picked this tomato; therefore it probably tastes great." Yes, this is an inductive argument, but the conclusion is not really a subjective judgment. Here, implicitly the speaker hasn't yet tasted the tomato, and is merely saying that if he does taste it probably it will taste great. if the speaker said, "i just picked this tomato and am tasting it; therefore it probably tastes great" we would have no idea what he or she meant.

DEDUCTION, INDUCTION, AND UNSTATED PREMISES

Somebody announces, "Rain is on its way." Somebody else asks how he knows. He says, "There's a south wind." Is the speaker trying to *demonstrate* rain is coming? Probably not. His thinking, spelled out, is probably something like this:

Stated premise: The wind is from the south.
Unstated premise: Around here, south winds are usually followed by rain.
Conclusion: There will be rain.

In other words, the speaker was merely trying to show that rain was a good possibility.

Notice, though, that the unstated premise in the argument could have been a universal statement to the effect that a south wind *always* is followed by rain at this particular location, in which case the argument would be deductive:

Stated premise: The wind is from the south.
Unstated premise: Around here, a south wind is always followed by rain.
Conclusion: There will be rain.

Spelled out this way, the speaker's thinking is deductive: It isn't possible for the premises to be true and the conclusion to be false. So one might wonder abstractly what the speaker intended—an inductive argument that supports the belief that rain is coming, or a deductive demonstration.

There is, perhaps, no way to be certain short of asking the speaker something like, "Are you 100 percent positive?" But experience ("background knowledge") tells us that wind from a particular direction is not a surefire indicator of rain. So probably the speaker did have in mind merely the first argument. He wasn't trying to present a 100 percent certain, knock-down demonstration that it would rain; he was merely trying to establish there was a good chance of rain.

You can always turn an inductive argument with an unstated premise into a deductively valid argument by supplying the right universal premise—a statement that something holds without exception or is true everywhere or in all cases. Is that what the speaker really has in mind, though? You have to use background knowledge and common sense to answer the question.

For example, you overhear someone saying,

Stacy and Justin are on the brink of divorce. They're always fighting.

One could turn this into a valid deductive argument by adding to it the universal statement "Every couple fighting is on the brink of divorce." But such an unqualified universal statement seems unlikely. Probably the speaker wasn't trying to demonstrate that Stacy and Justin are on the brink of divorce. He or she was merely trying to raise its likelihood. He or she was presenting evidence that Stacy and Justin are on the brink of divorce.

Often it is clear that the speaker does have a *deductive* argument in mind and has left some appropriate premise unstated. You overhear Professor Greene saying to Professor Brown, "Flunk her! This is the second time you've caught her cheating."

It would be strange to think that Professor Greene is merely trying to make it more likely that Professor Brown should flunk the student. Indeed, it is hard even to make sense of that suggestion. Professor Greene's argument, spelled out, must be this:

Stated premise: This is the second time you've caught her cheating.
Unstated premise: Anyone who has been caught cheating two times should be flunked.
Conclusion: She should be flunked.

So context and content often make it clear what unstated premise a speaker has in mind and whether the argument is deductive or inductive.

Unfortunately, though, this isn't always the case. We might hear someone say,

The bars are closed; therefore, it is later than 2 A.M.

If the unstated premise in the speaker's mind is something like "In this city, the bars all close at 2 A.M.," then presumably he or she is thinking deductively and is evidently proffering proof that it's after 2. But if the speaker's unstated premise is something like "Most bars in this city close at 2 A.M." or "Bars in this city usually close at 2 A.M.," then we have an inductive argument that merely supports the conclusion. So which is the unstated premise? We really can't say without knowing more about the situation or the speaker.

Is an Ad Photo an Argument?

Lars A. Niki

The short answer: no. The longer version: Still no. An advertising photograph can "give you a reason" for buying something only in the sense that it can *cause* you to think of a reason. A photo is not an argument.

The bottom line is this. Real-life arguments often leave a premise unstated. One such unstated premise might make the argument inductive; another might make it deductive. Usually, context or content make reasonably clear what is intended; other times they may not. When they don't, the best practice is to attribute to a speaker an unstated premise that at least is believable, everything considered. We'll talk about believability in Chapter 4.

BALANCE OF CONSIDERATIONS

Should I get a dog? Miss class to attend my cousin's wedding? Get chemo? Much everyday reasoning requires weighing considerations for and against thinking or doing something. Such reasoning, called **balance of considerations reasoning,** often contains both deductive and inductive elements. Here is an example:

> Should assault weapons be banned? On the one hand, doing that would violate the Second Amendment to the U.S. Constitution. But on the other hand, when guns were outlawed in Australia the number of accidental gun deaths fell dramatically; that would probably happen here, too. It is a tough call.

The first consideration mentioned in this passage—that banning assault weapons would violate the Second Amendment and therefore should not be done—is a deductive argument. The second consideration mentioned—that banning assault weapons would reduce the number of accidental gun deaths—is an inductive argument.

Inductive arguments can be compared as to strength and weakness; deductive arguments can be compared as to validity and soundness. Assigning weight to considerations can be difficult, of course, but it is not hopelessly arbitrary. In Chapter 12 of this book, we discuss the perspectives within which moral evaluations are made; you will see there that weighing considerations of the sort presented in the example above depend to a certain extent on the moral perspective one adheres to.

NOT PREMISES, CONCLUSIONS, OR ARGUMENTS

We hope you've noticed, when we use the word "argument," we are not talking about two people having a feud or fuss about something. That use of the word has nothing much to do with critical thinking, though many a heated exchange could use some critical thinking. Arguments in our sense do not even need two people; we make arguments for our own use all the time. And when we evaluate them, we think critically.

Speaking of what arguments are not, it's important to realize that not everything that might look like an argument, or like a premise or a conclusion, is one.

Selfies (and Other Pictures)

Pictures are not premises, conclusions, or arguments. Neither are movies. Your iPhone can do lots of things, but the jpegs stored on its SIM card aren't arguments. Sorry. Arguments have two parts, a premise part and a conclusion part, and both parts are propositional entities, which means (to repeat) that both parts must be expressible in declarative, true-or-false sentences. Movies and pictures can be moving, compelling, beautiful, complex, realistic, and so forth—but they cannot be either true or false. You can ask if what is depicted in a movie actually happened, or if the story upon which it is based is a true story, but you can't really ask if a movie itself is true or false—or valid or invalid or relatively strong or weak. Such questions don't make literal sense. If it doesn't make sense to think of a thing as true or false, it cannot be a premise or a conclusion. If it doesn't make sense to think of it as valid or invalid, or as being relatively strong or weak, it cannot be an argument.

The list of things that aren't premises or conclusions or arguments therefore also includes emotions, feelings, landscapes, faces, gestures, grunts, groans, bribes, threats, amusement parks, and hip-hop. Since they may *cause* you to have an opinion or to form a judgment about something or produce an argument, you might be tempted to think of them as premises, but causes are not premises. A cause isn't a propositional entity: it is neither true nor false. So it cannot be a premise.

If . . . then . . . Sentences

Sometimes sentences like the following are taken to state arguments:

> If you wash your car now, then it will get spots.

This statement might be the premise of an argument whose conclusion is "Therefore you shouldn't wash your car now." It might also be the conclusion of an argument whose premise is "It is raining." But though it *could* be a premise or a conclusion, it is not *itself* an argument. An argument has a premise and a conclusion, and, though the preceding statement has two parts, neither part by itself is either a premise or a conclusion. "If you wash your car now" is not a statement, and neither is "Then it will get spots." Neither of these phrases qualifies as either a premise or a conclusion. Bottom line: "If . . . then . . ." sentences are not arguments.

Lists of Facts

Though the following might look like an argument, it is nothing more than a list of facts:

> Identity theft is up at least tenfold over last year. More people have learned how easy it is to get hold of another's Social Security number, bank account numbers, and such. The local police department reminds everyone to keep close watch on who has access to such information.

Although they are related by being about the same subject, none of these claims is offered as a reason for believing another, and thus there is no argument here. But the following passage is different. See if you can spot why it is an argument:

> The number of people who have learned how to steal identities has doubled in the past year. So you are now more likely to become a victim of identity theft than you were a year ago.

Here, the first claim offers a reason for accepting the second claim; we now have an argument.

"A because B"

Sometimes the word "because" refers to the cause of something. But other times it refers to a premise of an argument. Mikey walks into the motel lobby, wearing a swimsuit and dripping wet. Consider these two statements:

"Mikey is in his swimsuit because he was swimming."

"Mikey was swimming because he's in his swimsuit."

These two sentences have the same form, "X because Y." But the sentence on the left *explains why* Mikey is wearing a swimsuit. The sentence on the right offers an argument *that* Mikey was swimming. Only the sentence on the right is an argument. Put it this way: What follows "because" in the sentence on the left is the *cause.* What follows "because" in the right-hand sentence is *evidence.*

Be sure you understand the difference between these two sentences. Arguments and cause-and-effect statements can both employ the phrase *"X because Y."* But there the similarity ends. When what follows "because" is a premise for accepting a contention, or evidence for it, we have an argument; when what follows "because" states the *cause* of something, we have a cause-and-effect explanation. These are entirely different enterprises. Arguing *that* a dog has fleas is different from explaining what *gave* it fleas. Arguing *that* violent crime has increased is different from explaining what *caused* it to increase.

ETHOS, PATHOS, AND LOGOS

When he was a young man, Alexander the Great conquered the world. Alexander was enormously proud of his accomplishment, and liked to name cities after himself. Alexander's teacher, the Greek philosopher Aristotle, had no cities named after him. There is no indication this disappointed Aristotle. And Aristotle's imprint on civilization turned out to be even more profound than Alexander's anyway.

Aristotle, who now is regarded as the father of logic, biology, and psychology, made enduring contributions to virtually every subject. These include (in addition to those just mentioned) physics, astronomy, meteorology, zoology, metaphysics, political science, economics, ethics, and rhetoric. When Aristotle is said to be the father of logic, it doesn't mean that Aristotle invented logic. It doesn't mean that before Aristotle came along, people were illogical and stupid. It only means that he was the first to study logic.

Among Aristotle's contributions in rhetoric was a theory of persuasion, which famously contained the idea that there are three modes by which a speaker may persuade an audience. Paraphrasing very loosely, Aristotle's idea was that we can be persuaded, first of all, by a speaker's personal attributes, including such things as his or her background, reputation, accomplishments, expertise, and similar things. Aristotle referred to this mode of persuasion as *ethos.* Second, a speaker can

■ first known pic of Aristotle taking a selfie.

persuade us by connecting with us on a personal level, and by arousing and appealing to our emotions by a skillful use of rhetoric. This mode of persuasion Aristotle termed *pathos.* And third, the speaker may persuade us by using information and arguments—what he called *logos.*

Unfortunately, logos—rational argumentation—is one of the least effective ways of winning someone to your point of view. That's why advertisers rarely bother with it. When the sellers of the first home automatic breadmaker found that its new kitchen device didn't interest people, they advertised the availability of a second model of the same machine, which was only slightly larger but much more expensive. When consumers saw that the first model was a great buy, they suddenly discovered they wanted one, and began snapping it up. Why try to persuade people by rational argument that they need a breadmaker when you can get them to think they do simply by making them believe they have sniffed out a bargain?*

Still, despite the general inefficacy of logos as a tool of persuasion, people do frequently use arguments when they try to persuade others. This might lead you to *define* an argument as an attempt to persuade. But that won't do. Remember, there are two kinds of argument. Deductive arguments are either sound or unsound, and whether a deductive argument is one or the other doesn't depend in the least on whether anyone is persuaded by it. Likewise, inductive arguments are in varying degrees strong or weak; their strength depends on the degree to which their premises elevate the probability of the conclusion, and that, too, is independent of whether anyone finds them persuasive. The very same argument might be persuasive to Parker but not to Moore, which shows that the persuasiveness of an argument is a subjective question of psychology, not of logic. Indeed, the individual who does *not* think critically is precisely the person who is persuaded by specious reasoning. People notoriously are unfazed by good arguments while finding even the worst arguments compelling. If you want to persuade people of something, try propaganda. Flattery has been known to work, too.

We will be looking at alternative modes of persuasion—what Aristotle called ethos and pathos—in considerable detail in Chapters 4, 5, 6, and 7. However, we do this not so you can persuade people, but so you can be alert to the influence of ethos and pathos on your own thinking.

Now, we aren't suggesting it is a bad thing to be a persuasive writer or speaker. Obviously it isn't; that's what rhetoric courses are for—to teach you to write persuasively. Let's just put it this way: Whenever you find yourself being persuaded by what someone says, find the "logos" in the "pathos," and be persuaded by it alone.

Exercise 2-1

The following exercises will give you practice (1) identifying premises and conclusions as well as words that indicate premises and conclusions, (2) telling the difference between deductive demonstrations and inductive supporting arguments, and (3) identifying balance of considerations arguments and inferences to the best explanation.

▲—See the answers section at the end of the chapter.

Indicate which blanks would ordinarily contain premises and which would ordinarily contain conclusions.

▲ 1. a , and b . Therefore, c .

▲ 2. a . So, since b , c .

*Dan Ariely, *Irrational Predictability* (new York: HarperCollins, 2008), 14, 15.